

Formulating cannabinoids with lipid excipients

An interactive quick guide



People make our name



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For more information:

Download our [brochures](#)

Access our dedicated webpage on [cannabinoid formulation](#)

Cannabinoids are highly soluble in lipid excipients

Saturation solubility of CBD and THC in lipid excipients (in mg/mL for liquid excipients; in mg/g for semi-solid and solid excipients).
A detailed method is available upon request.

Excipient	Chemical name	CBD (mg/mL or mg/g)	THC (mg/mL or mg/g)
Capryol® PGMC	Propylene glycol monocaprylate (Type I, monoesters >55%) NF	599	485
Capryol® 90	Propylene glycol monocaprylate (Type II, monoesters >90%) NF	548	496
Gelucire® 44/14	Lauroyl macrogol-32 glycerides EP Lauroyl polyoxyl-32 glycerides NF	>255	125
Gelucire® 48/16	PEG-32 stearate (type I) NF / EP pending	>300	>500
Gelucire® 50/13	Stearoyl macrogol-32 glycerides EP Stearoyl polyoxyl-32 glycerides NF	>400	>500
Labrafac™ Lipophile WL1349	Triglycerides medium-chain EP Medium-chain triglycerides NF Medium chain fatty acid triglyceride JPE	410	421
Labrafac™ PG	Propylene glycol dicaprylocaprate EP Propylene glycol dicaprylate/dicaprate NF	506	> 500
Labrafil® M 1944 CS	Oleoyl macrogol-6 glycerides EP Oleoyl polyoxyl-6 glycerides NF	421	454
Labrafil® M 2125 CS	Linoleoyl macrogol-6 glycerides EP Linoleoyl polyoxyl-6 glycerides NF	388	419
Labrasol® / Labrasol® ALF	Caprylocaproyl macrogol-8 glycerides EP Caprylocaproyl polyoxyl-8 glycerides NF	588	422
Lauroglycol™ FCC	Propylene glycol monolaurate (Type I, monoesters >45%) EP / NF	471	477
Lauroglycol™ 90	Propylene glycol monolaurate (Type II, monoesters >90%) EP / NF	539	470
Maisine® CC	Glycerol monolinoleate EP Glycerol monolinoleate NF	338	>500
Pecceol™	Glycerol monooleates (type 40) EP Glycerol monooleate (type 40) NF	318	487
Plurol® Oleique CC 497	Polyglyceryl-3 dioleate NF	274	>500
Transcutol® P / Transcutol® HP	Highly purified diethylene glycol monoethyl ether EP/NF	>530	434

USP descriptor for solubility:

- Very soluble >500 mg/mL
- Freely soluble 100-500 mg/mL

Oral drug delivery

Tinctures

Tinctures consist of oily solutions of a plant extract. Our oil-based excipients contribute to the bioavailability enhancement and palatability of the formulation. This is a flexible dosage form, suitable for different patient populations, including pediatric, geriatric and veterinary.

Gattefossé excipients	Functionalities						Market references / projects in development with cannabinoids	Regulatory information	
	Oily vehicle	Solubilizer	Increased bioavailability	Target lymphatic transport	Good dispersion in water	Taste		USP-NF	Ph. Eur.
Labrafac™ Lipophile WL 1349	✓	✓	✓			✓	Canada, Colombia	✓	✓
Labrafil® M 1944 CS	✓	✓	✓	✓	✓	✓	Canada	✓	✓
Labrafil® M 2125 CS	✓	✓	✓	✓	✓	✓	Canada	✓	✓
Maisine® CC	✓	✓	✓	✓		✓	Brazil, Canada, Colombia, USA, Paraguay	✓	✓
Peceol™	✓	✓	✓	✓		✓		✓	✓

For more information:

On [tinctures](#)

Capsules

Hard capsules consist of a gelatin or cellulose-based shell which can be filled with powder, liquid or semi-solid excipients. Soft capsules are usually filled with a liquid formulation, which can be an oily vehicle or a self-emulsifying system in order to enhance bioavailability. Capsules and tablets are the preferred pharmaceutical dosage form by the industry and patients, offering dosage uniformity.

Gattefossé excipients	Functionalities				Market references / projects in development with cannabinoids	Regulatory information	
	Oily vehicle	Solubilizer	Increased bioavailability	Target lymphatic transport		USP-NF	Ph. Eur.
<u>Capryol® 90</u>		✓	✓		Colombia	✓	
<u>Geloil™ SC</u>	✓	✓	✓			✓	
<u>Gelucire® 44/14</u>		✓	✓		Colombia, USA	✓	✓
<u>Gelucire® 48/16</u>		✓	✓		Colombia, Canada	✓	✓
<u>Labrafac™ Lipophile WL 1349</u>	✓	✓	✓		Canada	✓	✓
<u>Labrafil® M 1944 CS</u>	✓	✓	✓	✓		✓	✓
<u>Labrasol® ALF</u>		✓	✓		Canada, USA	✓	✓
<u>Lauroglycol™ 90</u>		✓	✓			✓	✓
<u>Maisine® CC</u>	✓	✓	✓	✓	Canada, France, Paraguay, USA	✓	✓
<u>Peceol™</u>	✓	✓	✓	✓	Canada, USA	✓	✓
<u>Plurol® Oleique CC 497</u>		✓	✓	✓	USA	✓	

For more information:

[On capsule filling/molding process](#)

[On the precedence of use of our excipients in hard capsules and in soft capsules \(log-in required\)](#)

Tablets

In tablet production, a lubricant is necessary to decrease friction between particles and compression tools. Immediate or sustained-release tablets can be produced depending on the onset of action desired.

Tablets and capsules are the preferred pharmaceutical dosage form by the industry and patients, offering dosage uniformity.

Gattefossé excipients	Functionalities			Market references / projects in development with cannabinoids	Regulatory information	
	Lubricant	Sustained release agent	Bioavailability enhancer		USP-NF	Ph. Eur.
<u>Compritol® 888 ATO</u>	✓	✓		Canada, USA	✓	✓
<u>Compritol® HD 5</u>	✓				✓	
<u>Gelucire® 48/16</u>			✓		✓	✓
<u>Precirol® ATO 5</u>	✓	✓			✓	✓

For more information:

[On tablet production](#)

[On lubricants](#)

[On sustained release agents](#)

Dermal and transdermal drug delivery

Dermal and transdermal delivery are additional routes of administration for cannabinoids. Multiple dosage forms can be produced by combining emulsifiers and thickeners to enhance textures. Solubilizers and penetration enhancers optimize drug delivery and penetration.

Creams

Gattefossé excipients	Functionalities			Market references / projects in development with cannabinoids	Regulatory information	
	Emulsifier	Solubilizer	Penetration enhancer		USP-NF	Ph. Eur.
<u>Capryol® PGMC</u>		✓	✓		✓	
<u>Gelot™ 64</u>	✓			Canada, Colombia	✓	
<u>Lauroglycol™ FCC</u>		✓	✓		✓	✓
<u>Tefose® 63</u>	✓			USA	✓	
<u>Tefose® 1500</u>	✓				✓	
<u>Transcutol® P</u>		✓	✓	Canada, Colombia, Paraguay, USA	✓	✓

For more information:

[On emulsion preparation and emulsifiers](#)
[On how to improve sensorial experience](#)

Gels

Gattefossé excipients	Functionalities		Market references / projects in development with cannabinoids	Regulatory information	
	Solubilizer	Penetration enhancer		USP-NF	Ph. Eur.
<u>Labrasol®</u>	✓	✓		✓	✓
<u>Lauroglycol™ FCC</u>	✓	✓		✓	✓
<u>Transcutol® P</u>	✓	✓	Canada, Colombia, UK, USA	✓	✓

For more information:

[On gel preparation](#)

Solutions/Sprays

Gattefossé excipients	Functionalities		Market references / projects in development with cannabinoids	Regulatory information	
	Solubilizer	Penetration enhancer		USP-NF	Ph. Eur.
<u>Capryol® PGMC</u>	✓	✓		✓	
<u>Labrafil® M 1944 CS</u>	✓			✓	✓
<u>Lauroglycol™ FCC</u>	✓	✓		✓	✓
<u>Labrasol®</u>	✓	✓	Canada	✓	✓
<u>Transcutol® P</u>	✓	✓		✓	✓

For more information:

[On microemulsion preparation](#)

Patches

Gattefossé excipients	Functionalities		Market references / projects in development with cannabinoids	Regulatory information	
	Solubilizer	Penetration enhancer		USP-NF	Ph. Eur.
<u>Capryol® PGMC</u>	✓	✓		✓	
<u>Labrasol®</u>	✓	✓	USA	✓	✓
<u>Lauroglycol™ FCC</u>	✓	✓		✓	✓
<u>Plurol® Oleique CC 497</u>	✓	✓		✓	
<u>Transcutol® P</u>	✓	✓	USA	✓	✓

Mucosal drug delivery

Mucosal delivery is an additional route of administration for cannabinoids avoiding the first pass metabolism and enabling fast delivery. Multiple dosage forms can be produced by combining emulsifiers and thickeners to enhance textures. Solubilizers and permeation enhancers optimize drug delivery and penetration.

Rectal drug delivery

Gattefossé excipients	Functionalities		Market references / projects in development with cannabinoids	Regulatory information	
	Solubilizer	Vehicle		USP-NF	Ph. Eur.
<u>Suppocire® CM</u>	✓	✓		✓	✓
<u>Suppocire® NB</u>	✓	✓		✓	✓
<u>Suppocire® AS2X</u>	✓	✓	Canada	✓	✓
<u>Suppocire® AGP</u>	✓	✓		✓	✓

Vaginal drug delivery

Gattefossé excipients	Functionalities			Market references / projects in development with cannabinoids	Regulatory information	
	Emulsifier	Solubilizer	Vehicle		USP-NF	Ph. Eur.
<u>Suppocire® NAS 50</u>		✓	✓		✓	✓
<u>Ovucire® 3460</u>		✓	✓	Canada		
<u>Tefose® 63</u>	✓				✓	
<u>Labrafil® M 1944 CS</u>		✓			✓	✓

For more information:

On suppository and pessary molding

Nasal drug delivery

Solutions/gels

Gattefossé excipients	Functionalities		Market references / projects in development with cannabinoids	Regulatory information	
	Solubilizer	Penetration enhancer		USP-NF	Ph. Eur.
<u>Capryol® PGMC</u>	✓	✓		✓	
<u>Labrafil® M 1944 CS</u>	✓	✓		✓	✓
<u>Labrasol®</u>	✓	✓		✓	✓
<u>Lauroglycol™ FCC</u>	✓	✓		✓	✓
<u>Transcutol® HP</u>	✓	✓	Colombia	✓	✓

Oromucosal drug delivery

Gattefossé excipients	Functionalities							Market references / projects in development with cannabinoids	Regulatory information	
	Oily vehicle	Solubilizer	Increased bioavailability	Penetration enhancer	Target lymphatic transport	Good dispersion in water	Taste		USP-NF	Ph. Eur.
<u>Capryol® 90</u>		✓		✓					✓	
<u>Labrafac™ Lipophile WL 1349</u>	✓	✓	✓				✓		✓	✓
<u>Labrafil® M 1944 CS</u>	✓	✓	✓		✓	✓	✓		✓	✓
<u>Labrafil® M 2125 CS</u>	✓	✓	✓		✓	✓	✓		✓	✓
<u>Labrasol®</u>		✓		✓						
<u>Lauroglycol™ 90</u>		✓		✓					✓	✓
<u>Maisine® CC</u>	✓	✓	✓		✓		✓	Canada, Colombia	✓	✓
<u>Peceol™</u>	✓				✓		✓			
<u>Transcutol® HP</u>		✓		✓				Canada	✓	✓

Edibles

Check local regulatory status of the ingredients.

Beverages

Gattefossé excipients	Functionalities						Market references / projects in development with cannabinoids	Regulatory information
	Oily vehicle	Solubilizer	Increased bioavailability	Target lymphatic transport	Good dispersion in water	Taste		GRAS status
<u>Labrafac™ Lipophile WL 1349</u>	✓	✓				✓	USA	
<u>Labrafil® M 1944 CS</u>	✓	✓	✓	✓	✓	✓		
<u>Labrafil® M 2125 CS</u>	✓	✓	✓	✓	✓	✓		
<u>Maisine® CC</u>	✓	✓	✓	✓		✓	Canada, USA	✓
<u>Peceol™</u>	✓	✓	✓	✓		✓		✓

Chewing gums / Lozenges

Gattefossé excipients	Functionalities						Market references / projects in development with cannabinoids	Regulatory information
	Oily vehicle	Solubilizer	Increased bioavailability	Target lymphatic transport	Enhanced permeation	Taste		GRAS status
<u>Labrafil® M 1944 CS</u>	✓	✓	✓	✓	✓	✓		
<u>Labrafil® M 2125 CS</u>	✓	✓	✓	✓	✓	✓		
<u>Labrasol® ALF</u>		✓	✓		✓		Europe	
<u>Maisine® CC</u>	✓	✓	✓	✓		✓	USA	✓

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